

Exploring The “Pandas” Library In Python

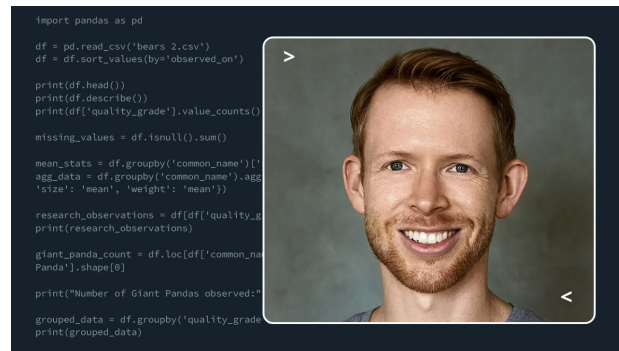
Omar Moustafa 900222400 & Mohamed Amin 900191168

CSCE 3101 - Term Project

July 22, 2024

Introduction to the Library

- Who & When:
 - Wes Mckinney
 - 2008
- Motivation:
 - To remove Python's lack of data structures that handle tabular data
- The name “*Pandas*” comes from:
 - ‘Panel Data’
 - ‘Python Data Analysis’
- Solves Data Analysis & Manipulation problems, most notably:
 - I/O tools
 - Data Cleansing
 - Data & Time Series Analysis



https://posit.co/wp-content/uploads/2023/11/Posit-Wes-McKinney_v3.jpg



<https://www.activestate.com/resources/quick-reads/what-is-pandas-in-python-everything-you-need-to-know/>

Introduction to the Library (cont.)

- What domains implement the use of Pandas?
 - Data Science
 - Cleaning data & preprocessing procedures
 - Healthcare
 - Processing medical records & analyzing healthcare analytics
 - Finance
 - Creating financial models & making quantitative analyses



<https://successive.cloud/wp-content/uploads/2022/04/Data-Analytics-Benefits.png>

Software Systems Built Using the Library

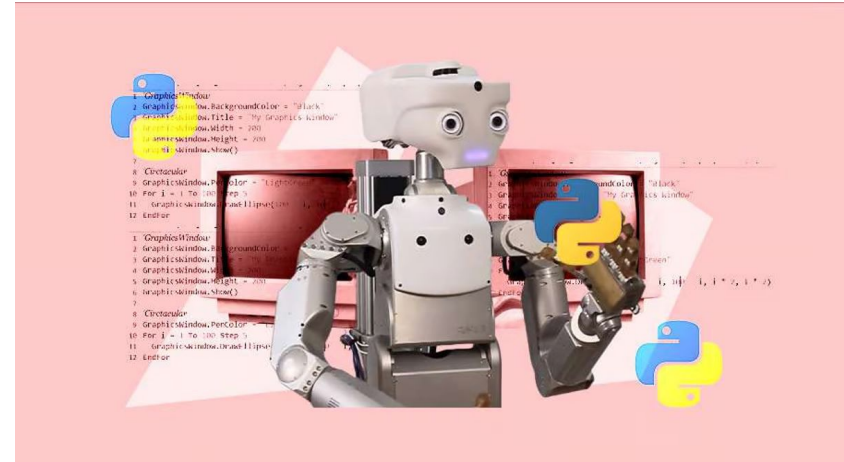
- Software #1 - Spotify's recommendation system
- Pandas contributes to such through:
 - 1. Data Transformation:
 - Analyzing data containing user activities, including listened-to-songs, playlists made, & user preferences
 - 2. Model Training & Evaluation:
 - Handling missing values, dividing datasets, and normalizing data become easier



<https://i.ytimg.com/vi/tooddaC14q4/sddefault.jpg>

Software Systems Built Using the Library (cont.)

- Software #2 - Robotic Software Systems
- For Robotic Software Systems to function well, excellent data processing and analysis are necessary
- Therefore, Pandas are used to/for:
 - 1. Sensor Data
 - 2. Path Planning



<https://www.analyticsinsight.net/latest-news/python-robotics-a-comprehensive-programming-guide>

Major Features of the Library

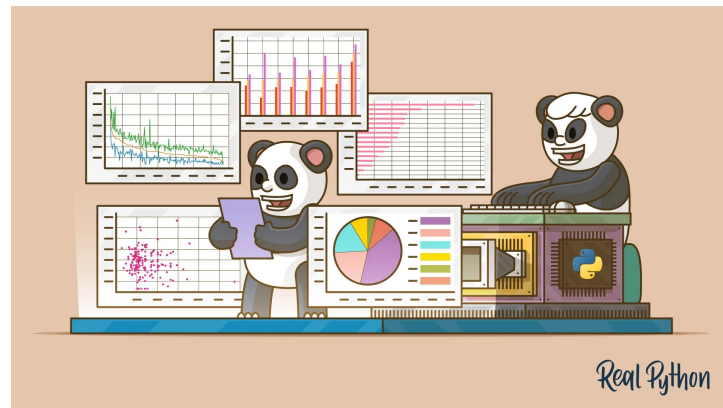
- Key Features & Capabilities:
 - 1. Importing Data
 - `read_csv()`
 - 2. Data Exploration
 - `head()`, `tail()`, `shape()`, `describe()`, `info()`
 - `iloc[row_number], [column_number], [['column1', 'column2']]`
 - 3. Data Cleansing
 - `isnull()`, `dropna()`, `fillna(value)`
 - 4. Data Analysis
 - `groupby()`
 - `sort_values()`, `sort_index()`
 - `query()`



https://favtutor.com/resources/images/uploads/mceu_16841658121636696850726.png

Major Features of the Library (cont.)

- The Standout Advantages:
 - 1. Ease of Use
 - Syntax is simple for beginners
 - Extensive documentation & tutorials
 - 2. Versatility of Data Structures
 - Offers the 1D “Series” & the 2D “DataFrame”
 - 3. Integration with the Python Ecosystem
 - Compatible with many other libraries (Matplotlib, NumPy, etc.)



https://files.realpython.com/media/How-to-Plot-With-Pandas_Watermarked.5bb0299e061b.jpg

Comparison with Similar Libraries



1. Dask, Vaex, & cuDF:

- Different Pandas APIs for out-of-memory computations
- cuDF is part of Nvidia's RAPIDS tools that use Nvidia GPUs for GPU computational analyses
- Vaex could analyze billions of rows per second
- Vaex was developed by astrophysicists who after 20 years of collecting stars data, they could not visualize it due to its complexity

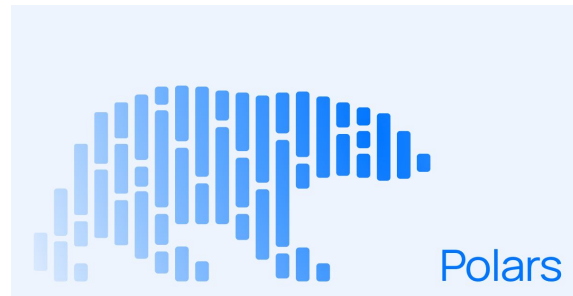


Comparison with Similar Libraries (cont.)

1. Dask:

- **Functionality:** Using distributed systems and parallel computing, it can process data that doesn't fit into memory by breaking it into smaller chunks and computing them in parallel.
- **Ease of use:** API is similar to Pandas but has a steeper learning curve.
- **Performance:** Pandas for small datasets and Dask for Large datasets.
- **Community:** Dask has a relatively large community but still smaller than Pandas.

Comparison with Similar Libraries (cont.)



2. Polars:

- Similar to Pandas but the focus is on optimizing performance
- **Functionality:** Polars utilizes parallel execution and lazy evaluation to enhance performance with large datasets
- **Ease of use:** API is similar to the dataframe API Pandas is known for but with its own syntax and set of conventions
- **Performance:** Polars supposedly beats Pandas in performance on similar datasets
- **Community:** Pandas still excel in its huge community and wide resources

Demonstration of Library Usage

<https://colab.research.google.com/drive/1Uvw7wcNTx0YUEf1wfv1OuXZNS5RSxAeX?usp=sharing>

Contribution Listing

- Omar:
 - Introduction to the Library
 - Software Systems Built Using the Library
 - Major Features of the Library
- Mohamed:
 - Comparison with Similar Libraries
 - Demonstration of Library Usage

References

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